

Roman Korol

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CURRENT POSITION	University of Rochester , Rochester, New York Postdoctoral Fellow, 2023-2025
FUTURE POSITION	Université de Sherbrooke , Sherbrooke, Quebec, Canada Assistant Professor, starting in January of 2026
EDUCATION	California Institute of Technology , Pasadena, California Ph.D. in Theoretical Chemistry, 2018–2023 University of Toronto , Toronto, Canada B.Sc. (Honours) in Chemistry with High Distinction, 2014–2018
FELLOWSHIPS AND AWARDS	Steadman Award 2024 Student Leader Award 2024 Gray-Hill Award Lecture 2023 Patricia Beckman Graduate Fellowship 2018 Michael Rebryk Memorial Scholarship 2018 Ivan Szak Scholarship in Chemistry 2018 St. Michael's College Silver Medal 2018 University of Toronto Excellence Award 2018 St. Michael's College In-Course Scholarship 2018 Canadian Society for Chemistry Silver medal 2018 CQIQC Undergraduate Summer Reaserch Program 2017 University of Toronto Excellence Award 2017 F. E. Beamish Scholarship in Chemistry 2017 Buduchnist Credit Union Scholarship 2017 Ivan Szak Scholarship in Chemistry 2017 Michael Both Award for Outstanding Commitment to Dance 2017 John Melady Memorial Scholarship 2017 C. W. Burton In-Course Scholarship 2017 Gollop Memorial Undergraduate Scholarship in Chemistry 2017 Dean's List Scholar 2017 University of Toronto Excellence Award 2016 Kupcinet-Getz Research Scholarship 2015 University of Toronto Mississauga Honour Roll 2015 Erindale Admission Scholarship 2014 Scholarship of the President of Ukraine 2014 (<i>awarded annually to approximately 250 highest achieving high school students out of over a million</i>) Lutsk's student of the year 2014

(awarded annually to the highest achieving high school graduate out of approximately 10,000)

First Prize at Intel-Eco Ukraine 2014, the national stage of [Intel ISEF](#) 2014

Gold medal at the International Ecology Project Olympiad 2013

PUBLICATIONS

14. Tiwari, V; Korol, R; Franco*, I Robust Purely Optical Signatures of Floquet States in Laser-Dressed Crystals. *Phys. Rev. Lett.* **2025**, 135 (18), 186901. DOI: [10.1103/5ywx-7db5](https://doi.org/10.1103/5ywx-7db5)
13. Korol*, R; Chen, X; Franco*, I High-Frequency Tails in Spectral Densities. *J. Phys. Chem. A* **2025**, 129 (15), 3587-3596. DOI: [10.1021/acs.jpca.5c00943](https://doi.org/10.1021/acs.jpca.5c00943)
12. Turner*, A C; Korol, R; Bill, M; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 2: Experimental determination of hydrogen isotopic equilibrium for ethane-H₂ from 30 to 200 °C and propane-H₂ from 75 to 200 °C. *Geochimica et Cosmochimica Acta* **2025**, 396, 91-106. DOI: [10.1016/j.gca.2025.02.033](https://doi.org/10.1016/j.gca.2025.02.033)
11. Korol*, R; Turner, A C; Nandi, A; Bowman, J M; Goddard, W A; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 1: Path-integral calculations with CCSD(T) quality potentials. *Geochimica et Cosmochimica Acta* **2025**, 396, 71-90. DOI: [10.1016/j.gca.2025.02.028](https://doi.org/10.1016/j.gca.2025.02.028)
10. Turner, A C; Korol, R; Eldridge, D L; Bill, M; Conrad, M E; Miller, T F; Stolper*, D A Experimental and theoretical determinations of hydrogen isotopic equilibrium in the system CH₄-H₂-H₂O from 3 to 200 °C. *Geochimica et Cosmochimica Acta* **2021**, 314, 223-269. DOI: [10.1016/j.gca.2021.04.026](https://doi.org/10.1016/j.gca.2021.04.026)
9. Korol, R; Rosa-Raíces, J L; Bou-Rabee, N; Miller*, T F Dimension-free path-integral molecular dynamics without preconditioning. *J. Chem. Phys.* **2020**, 152 (10), 104102. DOI: [10.1063/1.5134810](https://doi.org/10.1063/1.5134810)
8. Eldridge, D L; Korol, R; Lloyd, M K; Turner, A C; Webb, M A; Miller, T F; Stolper*, D A Comparison of Experimental vs Theoretical Abundances of ¹³CHD and ¹²CHD for Isotopically Equilibrated Systems from 1 to 500 °C. *ACS Earth Space Chem.* **2019**, 3 (12), 2747-2764. DOI: [10.1021/acsearthspacechem.9b00244](https://doi.org/10.1021/acsearthspacechem.9b00244)
7. Korol, R; Bou-Rabee, N; Miller*, T F Cayley modification for strongly stable path-integral and ring-polymer molecular dynamics. *J. Chem. Phys.* **2019**, 151 (12), 124103. DOI: [10.1063/1.5120282](https://doi.org/10.1063/1.5120282)
6. Korol, R; Segal*, D Machine Learning Prediction of DNA Charge Transport. *J. Phys. Chem. B* **2019**, 123 (13), 2801-2811. DOI: [10.1021/acs.jpcc.8b12557](https://doi.org/10.1021/acs.jpcc.8b12557)
5. Korol, R; Segal*, D From Exhaustive Simulations to Key Principles in DNA Nanoelectronics. *J. Phys. Chem. C* **2018**, 122 (8), 4206-4216. DOI: [10.1021/acs.jpcc.7b12744](https://doi.org/10.1021/acs.jpcc.7b12744)
4. Korol, R; Kilgour, M; Segal*, D ProbeZT: Simulation of transport coefficients of molecular electronic junctions under environmental effects using Büttiker's probes. *Computer Physics Communications* **2018**, 224, 396-404. DOI: [10.1016/j.cpc.2017.10.005](https://doi.org/10.1016/j.cpc.2017.10.005)
3. Korol, R; Kilgour, M; Segal*, D Thermopower of molecular junctions: Tunneling to hopping crossover in DNA. *J. Chem. Phys.* **2016**, 145 (22), 224702. DOI: [10.1063/1.4971167](https://doi.org/10.1063/1.4971167)
2. Longobardi, L E; Zatsepin, P; Korol, R; Liu, L; Grimme, S; Stephan*, D W Reactions of Boron-Derived Radicals with Nucleophiles. *J. Am. Chem. Soc.* **2016**, 139 (1), 426-435. DOI: [10.1021/jacs.6b11190](https://doi.org/10.1021/jacs.6b11190)
1. Korol*, R V; Yanchuk, O M; Marchuk, O V; Orlov, V F; Moroz, I A; Vyshnevskiy, O A Size Stabilizers in Two-electrode Synthesis of ZnO Nanorods. *Phys. Chem. Solid St.* **2021**, 22 (2), 380-387. DOI: [10.15330/pcss.22.2.380-387](https://doi.org/10.15330/pcss.22.2.380-387)

PRESENTATIONS AND AWARDS	American Conference on Theoretical Chemistry, Chapel Hill, North Carolina	2024
	Poster: “Analog Simulation of Open Quantum Dynamics”	
	Gray-Hill lecture at the Occidental college, Los Angeles, California.	2023
	<u>Award talk</u> : “A window to Earth’s past with the help of theoretical chemistry”	
	Canadian Chemistry Conference and Exhibition , Calgary, Canada	2022
	Contributed talk: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Molecular Science Mini-meeting , Montreal, Canada	2022
	Poster: “Dimension-free ring-polymer molecular dynamics”	
	ACS Spring meeting , San Diego, California	2022
	Poster: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Geological and Planetary Sciences seminar at Caltech, Pasadena, California	2022
	<u>Invited talk</u> : “ D and ^{13}C exchange equilibria using Path-Integral Monte-Carlo”	
	Berkeley Statistical Mechanics Meeting	2020
	Poster: “Cayley modification for strongly stable path-integral molecular dynamics”	
	CECAM BioMolecular Electronics Conference , Madrid, Spain	2018
	Poster: “Principles of Charge Transport in DNA: from extensive simulations to neural networks”	
	28 th Canadian Symposium on Theoretical and Computational Chemistry, Windsor, Canada	2018
	<u>Poster prize</u> : “Charge transport in DNA: From comprehensive simulations to key principles”	
	100 th Canadian Chemistry Conference , Toronto, Canada	2017
	<u>Poster prize</u> : “Tunneling to Hopping Crossover in Thermopower of DNA Molecular Junctions”	
	Chemical Biophysics Symposium , Toronto, Canada	2017
	Contributed talk: “DNA Molecular Junctions: Tunneling to Hopping Crossover”	
	33 rd Symposium on Chemical Physics, Waterloo, Canada	2017
	Contributed talk: “Probing mechanisms of charge transport in DNA with Landauer-Büttiker formalism”	
	45 th Southern Ontario Undergraduate Student Chemistry Conference , Toronto, Canada	2017
	<u>1st prize talk</u> : “Tunneling to Hopping Crossover in DNA & DNA-like molecular junctions”	
COMMUNITY VOLUNTEER INITIATIVES	Spearheaded and coordinated humanitarian supplies shipment to Ukraine from Caltech campus and beyond	2022–2023
	Volunteer at the Nova Ukraine non-profit, Stanford, California	2021–2023
	Website development, established and coordinated partnership with <i>Teach for Ukraine</i>	
	Volunteer at the Teach for Ukraine nonprofit, Kyiv, Ukraine	2021–2022
	Recruited and interviewed candidate teachers at the remote interview stage	
	International student orientation leader	2019, 2020
	“Big sibling” mentor for the incoming graduate students at Caltech	2019, 2020
	Science outreach program volunteer through Caltech Y	2018–2020
SERVICE	High-school tutoring with CAUSE Tutoring non-profit	2018–2019
	University of Toronto Peer Tutoring group tutor	2015–2018
	Student Representative at the Chemistry Department Advisory Committee	2016–2017
	2 nd year representative at the Chemistry student union	2016–2017
	Board member of the <i>Chemistry Connections</i> student group	2015–2016

TEACHING	<i>Course development:</i> Computational chemistry labs, Chem3 at Caltech Focus on structure-function relations and the dangers of approximations. High School Teacher, Rotman Arts and Science School, Vaughan, Canada Academic stream, grade 11 and 12 Chemistry, Grade 10 Science. Student placed 3rd in Vaughan, top 200 in Canada at the Avogadro chemistry contest International Chemistry Olympiad Coach Canadian National Team (4 students) – 2 bronze, 1 silver medals Ukrainian National team (1 student) – bronze medal Private tutoring of Chemistry, Physics and Math High school students: accepted to University college (UK), Columbia University (USA) and others Chemical Biology summer school, Lutsk, Ukraine Designed problems and experiments to help high school students master key concepts in chemistry	2022 2020–2022 Spring 2018 Spring 2014 2014–2018 Summer 2015
PEER REVIEW	ACS journals: ACS Physical Chemistry Au APS journals: Physical Review Letters, Physical Review A, B, and E Elsevier journals: Organic Geochemistry, Chemical Geology RSC journals: Physical chemistry chemical physics Wiley journals: Rapid communications in mass spectrometry	
SUMMER SCHOOLS AND WORKSHOPS	Condensed Phase Dynamics Workshop at TSRC (Virtual) Theoretical Chemistry School at TSRC , Telluride, Colorado Weizmann Institute of Science, Rehovot, Israel Kupcinet-Getz Scholar at Rubtchinski lab	2020 2019 2015
EMPLOYMENT	High School Teacher, Rotman Arts and Science School, Vaughan, Canada Research Assistant, Department of Linguistics, University of Toronto Heritage language variation and change project	2020–2022 2016–2018
EXTRA CURRICULARS	Rock climbing, weightlifting Guitar, base Ukrainian folk dance	since 2014 since 2012 since 2004
LANGUAGES	Fluent in Ukrainian, English & Russian	
COMPUTER LANGUAGES	Python, C++, MATLAB, FORTRAN, Mathematica, Bash; Web development (PHP & django, HTML, CSS, JS)	